Appendix for "Generalized and Unified Power Flow Framework for Converter-Dominated AC/DC Power Grids"

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APPENDIX A
THREE-WINDING TRANSFORMER EQUIVALENT MODEL

To incorporate a three-winding transformer into the power flow formulation, Section II-D1 of the paper represents it by an equivalent model of three two-winding transformers connected at an auxiliary bus. This equivalent model is required because each winding enters $\mathbf{Y}_{Bus}$ in (1) of the paper as a branch connecting two buses, whereas a three-winding transformer connects three. This section presents how the admittance of each winding is obtained and incorporated into $\mathbf{Y}_{Bus}$, along with the additional unknowns introduced by the auxiliary bus.

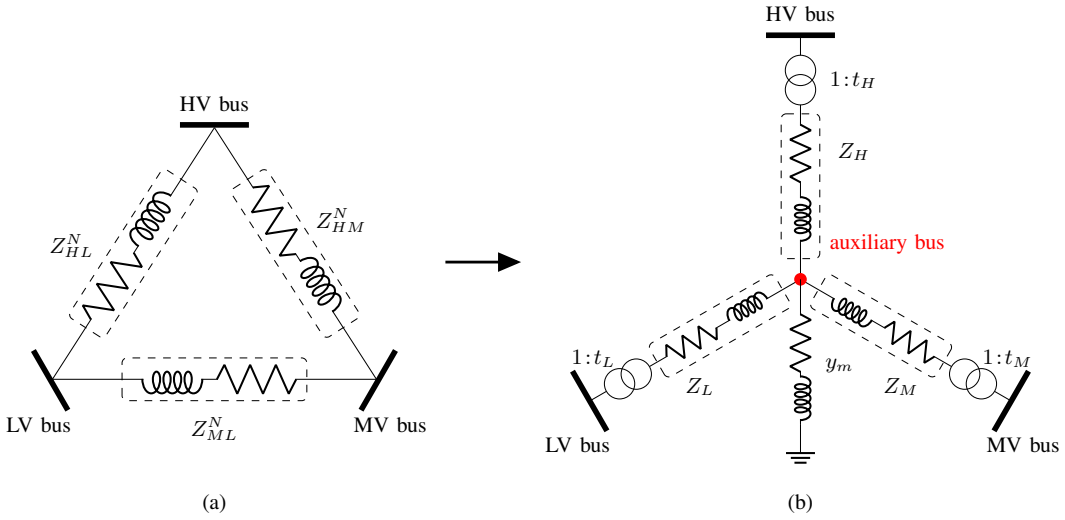


Fig. A1. Equivalent model of a three-winding transformer: (a) three-winding transformer with its short-circuit impedances and (b) star equivalent at the auxiliary bus.

Fig. A1(a) shows the three-winding transformer as it is typically represented, with its three short-circuit impedances Z^N_{HM}, Z^N_{ML}, and Z^N_{HL}, each consisting of a resistance and a reactance. Each of the three impedances corresponds to two of the HV, MV, and LV windings, and the superscript N denotes the value in per unit on the smaller of the two ratings. The three short-circuit impedances are then scaled to the system base S_{sys} as:

$$\begin{aligned} Z_{HM} &= Z^N_{HM}\frac{S_{sys}}{\min(S_H, S_M)} \\ Z_{ML} &= Z^N_{ML}\frac{S_{sys}}{\min(S_M, S_L)} \\ Z_{HL} &= Z^N_{HL}\frac{S_{sys}}{\min(S_H, S_L)} \end{aligned} \tag{A1}$$

where S_H, S_M, and S_L are the power ratings of the HV, MV, and LV windings. Fig. A1(b) shows the three-winding transformer represented by three two-winding transformers, each consisting of an ideal transformer and an impedance, with the HV, MV, and LV winding branches forming a star at the auxiliary bus. The impedances Z_H, Z_M, and Z_L of the winding branches are then obtained from Z_{HM}, Z_{ML}, and Z_{HL} in (A1) [1] as:

$$\begin{aligned} Z_H &= \tfrac{1}{2}\left(Z_{HM} + Z_{HL} - Z_{ML}\right) \\ Z_M &= \tfrac{1}{2}\left(Z_{HM} + Z_{ML} - Z_{HL}\right) \\ Z_L &= \tfrac{1}{2}\left(Z_{HL} + Z_{ML} - Z_{HM}\right). \end{aligned} \tag{A2}$$

In addition, the magnetizing branch of the three-winding transformer is modeled as the admittance y_m shunt-connected at the auxiliary bus.

Each winding branch enters $\mathbf{Y}_{Bus}$ as a two-winding transformer branch. In (21) of the paper, y_s is the inverse of the corresponding impedance in (A2), B_{ft} is zero because a winding branch has no line charging, and the tap ratio and phase shift of each winding are combined into a complex tap ratio as:

$$\begin{aligned} t_H &= \tau_H e^{\mathrm{j}\theta_H} \\ t_M &= \tau_M e^{\mathrm{j}\theta_M} \\ t_L &= \tau_L e^{\mathrm{j}\theta_L} \end{aligned} \tag{A3}$$

where t_H, t_M, and t_L are the complex tap ratios of the HV, MV, and LV windings, respectively, and τ and θ are the tap ratio and phase shift of each winding, corresponding to τ and θ_{shift} in (21) of the paper. For the phase shifts, the HV winding is taken as the reference, so $\theta_H = 0$ while θ_M and θ_L are the shifts of the MV and LV windings. The tap ratio and phase shift in (21) of the paper are assigned to bus f, which for a winding branch is its HV, MV, or LV bus rather than the auxiliary bus. The admittance y_m of the auxiliary bus enters (1) of the paper as the shunt admittance y_{sh}.

Since the auxiliary bus is included in $\mathbf{Y}_{Bus}$ as an AC bus, its voltage magnitude and phase angle enter $|\mathbf{V}_{AC}|$ and $\boldsymbol{\delta}$ in (24) of the paper as unknowns. A three-winding transformer is therefore represented by three two-winding transformers and one AC bus, and enters the unified formulation of the paper.

APPENDIX B
DERIVATION OF THE JACOBIAN BLOCKS

A. Overview of the Jacobian Structure

The augmented Jacobian in (41) of the paper consists of blocks whose elements are partial derivatives, and this section presents the derivation of every block. To identify each block, the unified state vector (39) and the unified mismatch vector (40) of the paper are expressed as:

$$\mathbf{x} = \big[\ \underbrace{|\mathbf{V}_{AC}|,\ \omega,\ \boldsymbol{\delta},\ \mathbf{V}_{DC}}_{\mathbf{x}_{base}},\ \underbrace{\tilde{\mathbf{P}}_{G,AC},\ \tilde{\mathbf{Q}}_{G,AC},\ \tilde{\mathbf{P}}_{G,DC}}_{\mathbf{x}_G},\ \underbrace{\tilde{\mathbf{P}}_{IC},\ \tilde{\mathbf{Q}}_{IC}}_{\mathbf{x}_{IC}}\ \big]^T \tag{B1}$$

$$\mathbf{F} = \big[\ \underbrace{\Delta\mathbf{P}_{AC},\ \Delta\mathbf{Q}_{AC},\ \Delta\mathbf{P}_{DC}}_{\mathbf{F}_{base}},\ \underbrace{\mathbf{F}_{P,AC},\ \mathbf{F}_{Q,AC},\ \mathbf{F}_{P,DC}}_{\mathbf{F}_G},\ \underbrace{\mathbf{F}_{P,IC},\ \mathbf{F}_{Q,IC}}_{\mathbf{F}_{IC}}\ \big]^T \tag{B2}$$

where the baseline groups $\mathbf{x}_{base}$ and $\mathbf{F}_{base}$ follow (24) and (30) of the paper, the generator groups $\mathbf{x}_G$ and $\mathbf{F}_G$ follow (32) and (34), and the IC groups $\mathbf{x}_{IC}$ and $\mathbf{F}_{IC}$ follow (35) and (38). Here $\Delta\mathbf{P}_{AC}$, $\Delta\mathbf{Q}_{AC}$, and $\Delta\mathbf{P}_{DC}$ denote the three rows of (30) of the paper.

Table B1 shows the augmented Jacobian, each block $\mathbf{J}_{a,b}$ corresponding to the ath group of (B2) and the bth group of (B1). Here a is the position of its mismatch group among the eight groups of (B2), and b that of its state group among the nine groups of (B1).

TABLE B1
BLOCK STRUCTURE OF THE AUGMENTED JACOBIAN $\mathbf{J} = \partial\mathbf{F}/\partial\mathbf{x}$ WITH SHADING MARKING THE SECTION IN WHICH EACH BLOCK IS DERIVED (BLUE FOR APPENDIX B-B, GREEN FOR APPENDIX B-C1, AND ORANGE FOR APPENDIX B-C2)

	$\lvert\mathbf{V}_{AC}\rvert$	ω	$\boldsymbol{\delta}$	$\mathbf{V}_{DC}$	$\tilde{\mathbf{P}}_{G,AC}$	$\tilde{\mathbf{Q}}_{G,AC}$	$\tilde{\mathbf{P}}_{G,DC}$	$\tilde{\mathbf{P}}_{IC}$	$\tilde{\mathbf{Q}}_{IC}$
$\Delta\mathbf{P}_{AC}$	$\mathbf{J}_{1,1}$	$\mathbf{J}_{1,2}$	$\mathbf{J}_{1,3}$	$\mathbf{J}_{1,4}$	$\mathbf{J}_{1,5}$	$\mathbf{J}_{1,6}$	$\mathbf{J}_{1,7}$	$\mathbf{J}_{1,8}$	$\mathbf{J}_{1,9}$
$\Delta\mathbf{Q}_{AC}$	$\mathbf{J}_{2,1}$	$\mathbf{J}_{2,2}$	$\mathbf{J}_{2,3}$	$\mathbf{J}_{2,4}$	$\mathbf{J}_{2,5}$	$\mathbf{J}_{2,6}$	$\mathbf{J}_{2,7}$	$\mathbf{J}_{2,8}$	$\mathbf{J}_{2,9}$
$\Delta\mathbf{P}_{DC}$	$\mathbf{J}_{3,1}$	$\mathbf{J}_{3,2}$	$\mathbf{J}_{3,3}$	$\mathbf{J}_{3,4}$	$\mathbf{J}_{3,5}$	$\mathbf{J}_{3,6}$	$\mathbf{J}_{3,7}$	$\mathbf{J}_{3,8}$	$\mathbf{J}_{3,9}$
$\mathbf{F}_{P,AC}$	$\mathbf{J}_{4,1}$	$\mathbf{J}_{4,2}$	$\mathbf{J}_{4,3}$	$\mathbf{J}_{4,4}$	$\mathbf{J}_{4,5}$	$\mathbf{J}_{4,6}$	$\mathbf{J}_{4,7}$	$\mathbf{J}_{4,8}$	$\mathbf{J}_{4,9}$
$\mathbf{F}_{Q,AC}$	$\mathbf{J}_{5,1}$	$\mathbf{J}_{5,2}$	$\mathbf{J}_{5,3}$	$\mathbf{J}_{5,4}$	$\mathbf{J}_{5,5}$	$\mathbf{J}_{5,6}$	$\mathbf{J}_{5,7}$	$\mathbf{J}_{5,8}$	$\mathbf{J}_{5,9}$
$\mathbf{F}_{P,DC}$	$\mathbf{J}_{6,1}$	$\mathbf{J}_{6,2}$	$\mathbf{J}_{6,3}$	$\mathbf{J}_{6,4}$	$\mathbf{J}_{6,5}$	$\mathbf{J}_{6,6}$	$\mathbf{J}_{6,7}$	$\mathbf{J}_{6,8}$	$\mathbf{J}_{6,9}$
$\mathbf{F}_{P,IC}$	$\mathbf{J}_{7,1}$	$\mathbf{J}_{7,2}$	$\mathbf{J}_{7,3}$	$\mathbf{J}_{7,4}$	$\mathbf{J}_{7,5}$	$\mathbf{J}_{7,6}$	$\mathbf{J}_{7,7}$	$\mathbf{J}_{7,8}$	$\mathbf{J}_{7,9}$
$\mathbf{F}_{Q,IC}$	$\mathbf{J}_{8,1}$	$\mathbf{J}_{8,2}$	$\mathbf{J}_{8,3}$	$\mathbf{J}_{8,4}$	$\mathbf{J}_{8,5}$	$\mathbf{J}_{8,6}$	$\mathbf{J}_{8,7}$	$\mathbf{J}_{8,8}$	$\mathbf{J}_{8,9}$

In this section, i and j denote AC buses, k and l DC buses, c and c' ICs, g generators, λ loads, and d DC/DC converters. At each bus, $\mathcal{G}_i$, $\mathcal{L}_i$, and $\mathcal{C}_i$ denote the sets of generators, loads, and ICs connected to AC bus i, and $\mathcal{G}_k$, $\mathcal{L}_k$, and $\mathcal{C}_k$ those connected to DC bus k. For DC/DC converters, $\mathcal{D}_k^{HV}$ and $\mathcal{D}_k^{LV}$ denote the sets of converters whose high- and low-voltage bus is bus k, respectively.

As marked in Table B1, the following subsections derive every block explicitly. Appendix B-B treats the block $\partial\mathbf{F}_{base}/\partial\mathbf{x}_{base}$, and Appendix B-C the blocks introduced by the augmented variables. Appendix B-D then describes the structure of the augmented Jacobian, which changes according to the operating mode of each generator, IC, and DC/DC converter and to whether it reaches an output limit.

B. Baseline Blocks

The block $\partial\mathbf{F}_{base}/\partial\mathbf{x}_{base}$ of the augmented Jacobian in (41) of the paper corresponds to the network rows $\mathbf{F}_{base}$ and the state columns $\mathbf{x}_{base}$ of Table B1.

The partial derivatives of $\Delta\mathbf{P}_{AC}$ and $\Delta\mathbf{Q}_{AC}$ with respect to $|\mathbf{V}_{AC}|$ and $\boldsymbol{\delta}$ give $\mathbf{J}_{1,1}$, $\mathbf{J}_{2,1}$, $\mathbf{J}_{1,3}$, and $\mathbf{J}_{2,3}$, whose (i,j)th elements are defined as:

$$\mathbf{J}_{1,1}(i,j)=\begin{cases}-|V_i||Y_{ij}|\cos\gamma_{ij}, & j\neq i\\ -2|V_i||Y_{ii}|\cos\theta_{ii}-\sum\limits_{j\neq i}|V_j||Y_{ij}|\cos\gamma_{ij}-\sum\limits_{\lambda\in\mathcal{L}_i}P^0_{LAC,\lambda}\left(\dfrac{2Z_{p,\lambda}|V_i|}{|V_i^0|^2}+\dfrac{I_{p,\lambda}}{|V_i^0|}\right)\big[1+k_{pw,\lambda}(\omega-\omega^0)\big], & j=i\end{cases} \tag{B3}$$

$$\mathbf{J}_{2,1}(i,j)=\begin{cases}-|V_i||Y_{ij}|\sin\gamma_{ij}, & j\neq i\\ 2|V_i||Y_{ii}|\sin\theta_{ii}-\sum\limits_{j\neq i}|V_j||Y_{ij}|\sin\gamma_{ij}-\sum\limits_{\lambda\in\mathcal{L}_i}Q^0_{LAC,\lambda}\left(\dfrac{2Z_{q,\lambda}|V_i|}{|V_i^0|^2}+\dfrac{I_{q,\lambda}}{|V_i^0|}\right)\big[1+k_{qw,\lambda}(\omega-\omega^0)\big], & j=i\end{cases} \tag{B4}$$

$$\mathbf{J}_{1,3}(i,j)=\begin{cases}-|V_i||V_j||Y_{ij}|\sin\gamma_{ij}, & j\neq i\\ |V_i|\sum\limits_{j\neq i}|V_j||Y_{ij}|\sin\gamma_{ij}, & j=i\end{cases} \tag{B5}$$

$$\mathbf{J}_{2,3}(i,j)=\begin{cases}|V_i||V_j||Y_{ij}|\cos\gamma_{ij}, & j\neq i\\ -|V_i|\sum\limits_{j\neq i}|V_j||Y_{ij}|\cos\gamma_{ij}, & j=i\end{cases} \tag{B6}$$

where $\gamma_{ij}=\delta_i-\delta_j-\theta_{ij}$ is the angle appearing in the calculated injections (25) of the paper. Similarly, the partial derivatives of $\Delta\mathbf{P}_{AC}$ and $\Delta\mathbf{Q}_{AC}$ with respect to ω give $\mathbf{J}_{1,2}$ and $\mathbf{J}_{2,2}$, whose ith elements are defined as:

$$\begin{aligned}\mathbf{J}_{1,2}(i)&=-\sum_{\lambda\in\mathcal{L}_i}P^0_{LAC,\lambda}\Big[Z_{p,\lambda}\big(\tfrac{|V_i|}{|V_i^0|}\big)^2+I_{p,\lambda}\tfrac{|V_i|}{|V_i^0|}+P_{p,\lambda}\Big]k_{pw,\lambda}-|V_i|\sum_{j\in\mathcal{B}_{AC}}|V_j|\Big(\tfrac{\partial|Y_{ij}|}{\partial\omega}\cos\gamma_{ij}+|Y_{ij}|\tfrac{\partial\theta_{ij}}{\partial\omega}\sin\gamma_{ij}\Big),\\ \mathbf{J}_{2,2}(i)&=-\sum_{\lambda\in\mathcal{L}_i}Q^0_{LAC,\lambda}\Big[Z_{q,\lambda}\big(\tfrac{|V_i|}{|V_i^0|}\big)^2+I_{q,\lambda}\tfrac{|V_i|}{|V_i^0|}+Q_{q,\lambda}\Big]k_{qw,\lambda}-|V_i|\sum_{j\in\mathcal{B}_{AC}}|V_j|\Big(\tfrac{\partial|Y_{ij}|}{\partial\omega}\sin\gamma_{ij}-|Y_{ij}|\tfrac{\partial\theta_{ij}}{\partial\omega}\cos\gamma_{ij}\Big).\end{aligned} \tag{B7}$$

Specifically, these derivatives follow from the frequency dependence of Y_{ij} in (1) of the paper:

$$\begin{aligned}\frac{\partial Y_{ij}}{\partial\omega}&=\begin{cases}\sum\limits_{\substack{j\in\mathcal{B}_{AC}\\ j\neq i}}\Big(-\mathrm{j}X_{ij}y_{ij}^2+\mathrm{j}\tfrac{B_{ij}}{2}\Big)+\mathrm{j}\tfrac{\partial b_{sh,i}}{\partial\omega} & i=j\\ \mathrm{j}X_{ij}y_{ij}^2 & i\neq j\end{cases}\\ \frac{\partial|Y_{ij}|}{\partial\omega}&=\frac{\mathrm{Re}\{Y_{ij}^*\,\partial Y_{ij}/\partial\omega\}}{|Y_{ij}|},\qquad \frac{\partial\theta_{ij}}{\partial\omega}=\frac{\mathrm{Im}\{Y_{ij}^*\,\partial Y_{ij}/\partial\omega\}}{|Y_{ij}|^2}\end{aligned} \tag{B8}$$

where $y_{ij}=(R_{ij}+\mathrm{j}\omega X_{ij})^{-1}$, and R_{ij}, X_{ij}, and B_{ij} denote the resistance, reactance, and total line-charging susceptance of the branch between AC buses i and j at the nominal frequency, so that the reactance and susceptance in (1) of the paper are ωX_{ij} and ωB_{ij}. For a transformer branch from bus f to bus t, the terms of (B8) follow (21) of the paper, with y_s and B_{ft} in place of y_{ij} and B_{ij}. The term in the diagonal element of bus f is divided by τ^2, and the terms in the (f,t)th and (t,f)th elements are divided by $\tau e^{-\mathrm{j}\theta_{shift}}$ and $\tau e^{\mathrm{j}\theta_{shift}}$, respectively, whereas the term in the diagonal element of bus t is unchanged. The shunt susceptance $b_{sh,i}$ in (1) of the paper, including the filter susceptance $B_{f,c}$ of an IC, is $b^0_{sh,i}\omega$ when capacitive and $b^0_{sh,i}/\omega$ when inductive, where $b^0_{sh,i}$ is its value at the nominal frequency. Accordingly, $\partial b_{sh,i}/\partial\omega$ is $b^0_{sh,i}$ and $-b^0_{sh,i}/\omega^2$, respectively. The DC voltage does not appear in the specified injections (27)–(28) of the paper, where the augmented variables replace the IC terms:

$$\mathbf{J}_{1,4}=\mathbf{J}_{2,4}=\mathbf{0}. \tag{B9}$$

On the DC side, the partial derivatives of $\Delta\mathbf{P}_{DC}$ with respect to $\mathbf{V}_{DC}$ give $\mathbf{J}_{3,4}$, whose (k,l)th elements are defined as:

$$\mathbf{J}_{3,4}(k,l)=\begin{cases}-V_kG_{kl}+\Delta\mathbf{J}_{3,4}(k,l), & l\neq k\\ -\sum\limits_{\lambda\in\mathcal{L}_k}P^0_{LDC,\lambda}\Big(\dfrac{2Z_{p,\lambda}V_k}{(V_k^0)^2}+\dfrac{I_{p,\lambda}}{V_k^0}\Big)-2V_kG_{kk}-\sum\limits_{l\neq k}V_lG_{kl}+\Delta\mathbf{J}_{3,4}(k,k), & l=k\end{cases} \tag{B10}$$

The DC/DC converter terms of (22)–(23) of the paper introduce the additional term $\Delta\mathbf{J}_{3,4}$, whose (k,l)th elements are defined as:

$$\Delta\mathbf{J}_{3,4}(k,l)=\begin{cases}-\sum\limits_{d\in\mathcal{D}_k^{HV}}\dfrac{2m_{HV,d}}{V_{DC,\max,\mathrm{HV},d}-V_{DC,\min,\mathrm{HV},d}}-\sum\limits_{d\in\mathcal{D}_k^{LV}}\dfrac{2m_{LV,d}}{V_{DC,\max,\mathrm{LV},d}-V_{DC,\min,\mathrm{LV},d}}, & l=k\\ \sum\limits_{\substack{d\in\mathcal{D}_k^{LV}\\ d\in\mathcal{D}_l^{HV}}}\dfrac{2m_{HV,d}}{V_{DC,\max,\mathrm{HV},d}-V_{DC,\min,\mathrm{HV},d}}+\sum\limits_{\substack{d\in\mathcal{D}_k^{HV}\\ d\in\mathcal{D}_l^{LV}}}\dfrac{2m_{LV,d}}{V_{DC,\max,\mathrm{LV},d}-V_{DC,\min,\mathrm{LV},d}}, & l\neq k\end{cases} \tag{B11}$$

For each IC c, the conversion loss in (16) of the paper depends on the converter current in (17) of the paper. Its derivative with respect to a variable x of the unified state vector is therefore obtained by the chain rule as:

$$\begin{aligned}\frac{\partial P_{IC,loss,c}}{\partial x}&=\left(b_c+2c_c|I_{IC,c}|\right)\frac{\partial|I_{IC,c}|}{\partial x},\\ \frac{\partial|I_{IC,c}|}{\partial x}&=\begin{cases}-\dfrac{|I_{IC,c}|}{|V_{\mathrm{conv},c}|}, & x=|V_{\mathrm{conv},c}|\\ \dfrac{\tilde{P}_{IC,AC,c}}{|I_{IC,c}|\,|V_{\mathrm{conv},c}|^2}, & x=\tilde{P}_{IC,AC,c}\\ \dfrac{\tilde{Q}_{IC,AC,c}}{|I_{IC,c}|\,|V_{\mathrm{conv},c}|^2}, & x=\tilde{Q}_{IC,AC,c}\\ 0, & \text{otherwise.}\end{cases}\end{aligned} \tag{B12}$$

The partial derivative of the IC's DC-side injection (18) of the paper with respect to $|V_{\mathrm{conv},c}|$, obtained from (B12), gives $\mathbf{J}_{3,1}$, whose (k,i)th elements are defined as:

$$\mathbf{J}_{3,1}(k,i)=\sum_{\substack{c\in\mathcal{C}_k\\ c\in\mathcal{C}_i}}\left(b_c+2c_c|I_{IC,c}|\right)\frac{|I_{IC,c}|}{|V_{\mathrm{conv},c}|}. \tag{B13}$$

The DC mismatch $\Delta\mathbf{P}_{DC}$ contains neither the AC subgrid frequencies nor the phase angles, so the remaining blocks are zero:

$$\mathbf{J}_{3,2}=\mathbf{J}_{3,3}=\mathbf{0}. \tag{B14}$$

C. Augmented Blocks

The augmented Jacobian contains, apart from $\partial\mathbf{F}_{base}/\partial\mathbf{x}_{base}$, the blocks in the $\mathbf{x}_G$ and $\mathbf{x}_{IC}$ columns and the $\mathbf{F}_G$ and $\mathbf{F}_{IC}$ rows of Table B1. The following subsections treat the blocks of $\mathbf{x}_G$ and $\mathbf{F}_G$ for the droop deadband and the output limits, then those of $\mathbf{x}_{IC}$ and $\mathbf{F}_{IC}$ for the converter station losses and the output limits.

1) Generators: The augmented variables $\tilde{\mathbf{P}}_{G,AC}$, $\tilde{\mathbf{Q}}_{G,AC}$, and $\tilde{\mathbf{P}}_{G,DC}$ replace the generator outputs $P_{G,AC,g}$, $Q_{G,AC,g}$, and $P_{G,DC,g}$ in the specified injections (27)–(29), as in Section III-B2 of the paper. Each of these variables appears in an injection only if its generator is connected to that bus, so $\mathbf{J}_{1,5}$, $\mathbf{J}_{2,6}$, and $\mathbf{J}_{3,7}$ are incidence blocks, whose (i,g)th and (k,g)th elements are defined as:

$$\begin{aligned}\mathbf{J}_{1,5}(i,g)=\mathbf{J}_{2,6}(i,g)&=\begin{cases}1, & g\in\mathcal{G}_i\\ 0, & \text{otherwise,}\end{cases}\\ \mathbf{J}_{3,7}(k,g)&=\begin{cases}1, & g\in\mathcal{G}_k\\ 0, & \text{otherwise.}\end{cases}\end{aligned} \tag{B15}$$

Each injection contains only one of $\tilde{\mathbf{P}}_{G,AC}$, $\tilde{\mathbf{Q}}_{G,AC}$, and $\tilde{\mathbf{P}}_{G,DC}$, so the remaining blocks are zero:

$$\begin{aligned}\mathbf{J}_{1,6}=\mathbf{J}_{1,7}&=\mathbf{0},\\ \mathbf{J}_{2,5}=\mathbf{J}_{2,7}&=\mathbf{0},\\ \mathbf{J}_{3,5}=\mathbf{J}_{3,6}&=\mathbf{0}.\end{aligned} \tag{B16}$$

The partial derivatives of $\mathbf{F}_{P,AC}$, $\mathbf{F}_{Q,AC}$, and $\mathbf{F}_{P,DC}$ with respect to ω, $|\mathbf{V}_{AC}|$, and $\mathbf{V}_{DC}$ give $\mathbf{J}_{4,2}$, $\mathbf{J}_{5,1}$, and $\mathbf{J}_{6,4}$, whose gth, (g,i)th, and (g,k)th elements are defined as:

$$\mathbf{J}_{4,2}(g) = \begin{cases} m_{pAC,g}, & |\omega^0 - \omega| > \omega_{DB}, \\ & P^{\min}_{G,AC,g} \le P^0_{G,AC,g} + m_{pAC,g}(\omega^0 \mp \omega_{DB} - \omega) \le P^{\max}_{G,AC,g} \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{J}_{5,1}(g,i) = \begin{cases} m_{qAC,g}, & g \in \mathcal{G}_i, \quad \big||V_i^0| - |V_i|\big| > V_{DB}, \\ & Q^{\min}_{G,AC,g} \le Q^0_{G,AC,g} + m_{qAC,g}(|V_i^0| \mp V_{DB} - |V_i|) \le Q^{\max}_{G,AC,g} \\ 0, & \text{otherwise} \end{cases} \tag{B17}$$

$$\mathbf{J}_{6,4}(g,k) = \begin{cases} m_{pDC,g}, & g \in \mathcal{G}_k, \quad |V_k^0 - V_k| > V_{DC,DB}, \\ & P^{\min}_{G,DC,g} \le P^0_{G,DC,g} + m_{pDC,g}(V_k^0 \mp V_{DC,DB} - V_k) \le P^{\max}_{G,DC,g} \\ 0, & \text{otherwise.} \end{cases}$$

The elements are nonzero only where the state lies outside the deadband and the droop characteristic within the output limits, so the output is not fixed. The upper and lower signs of $\mp$ apply below and above the deadband, respectively, as in (33) of the paper. Each constraint contains only one of $|\mathbf{V}_{AC}|$, ω, $\boldsymbol{\delta}$, and $\mathbf{V}_{DC}$, so the remaining blocks are zero:

$$\begin{aligned} \mathbf{J}_{4,1} &= \mathbf{J}_{4,3} = \mathbf{J}_{4,4} = \mathbf{0}, \\ \mathbf{J}_{5,2} &= \mathbf{J}_{5,3} = \mathbf{J}_{5,4} = \mathbf{0}, \\ \mathbf{J}_{6,1} &= \mathbf{J}_{6,2} = \mathbf{J}_{6,3} = \mathbf{0}. \end{aligned} \tag{B18}$$

The three constraint groups $\mathbf{F}_{P,AC}$, $\mathbf{F}_{Q,AC}$, and $\mathbf{F}_{P,DC}$ contain $\tilde{\mathbf{P}}_{G,AC}$, $\tilde{\mathbf{Q}}_{G,AC}$, and $\tilde{\mathbf{P}}_{G,DC}$ themselves, and no remaining term of each constraint depends on its own variable. The blocks $\mathbf{J}_{4,5}$, $\mathbf{J}_{5,6}$, and $\mathbf{J}_{6,7}$ are therefore identity matrices:

$$\mathbf{J}_{4,5} = \mathbf{J}_{5,6} = \mathbf{J}_{6,7} = \mathbf{I}. \tag{B19}$$

Each constraint contains only one of $\tilde{\mathbf{P}}_{G,AC}$, $\tilde{\mathbf{Q}}_{G,AC}$, and $\tilde{\mathbf{P}}_{G,DC}$, so the remaining blocks are zero:

$$\begin{aligned} \mathbf{J}_{4,6} &= \mathbf{J}_{4,7} = \mathbf{0}, \\ \mathbf{J}_{5,5} &= \mathbf{J}_{5,7} = \mathbf{0}, \\ \mathbf{J}_{6,5} &= \mathbf{J}_{6,6} = \mathbf{0}. \end{aligned} \tag{B20}$$

Each constraint contains none of the IC variables $\tilde{\mathbf{P}}_{IC}$ and $\tilde{\mathbf{Q}}_{IC}$ either, so the corresponding blocks are zero:

$$\begin{aligned} \mathbf{J}_{4,8} &= \mathbf{J}_{4,9} = \mathbf{0}, \\ \mathbf{J}_{5,8} &= \mathbf{J}_{5,9} = \mathbf{0}, \\ \mathbf{J}_{6,8} &= \mathbf{J}_{6,9} = \mathbf{0}. \end{aligned} \tag{B21}$$

2) Interlinking Converters: The converter-bus injections $\tilde{\mathbf{P}}_{IC}$ and $\tilde{\mathbf{Q}}_{IC}$ replace the IC terms $P_{IC,AC,c}$ and $Q_{IC,AC,c}$ in the specified injections (27)–(28), as in Section III-B3 of the paper, so $\mathbf{J}_{1,8}$ and $\mathbf{J}_{2,9}$ are again incidence blocks, whose (i,c)th elements are defined as:

$$\mathbf{J}_{1,8}(i,c) = \mathbf{J}_{2,9}(i,c) = \begin{cases} 1, & c \in \mathcal{C}_i \\ 0, & \text{otherwise.} \end{cases} \tag{B22}$$

Each injection contains only one of $\tilde{\mathbf{P}}_{IC}$ and $\tilde{\mathbf{Q}}_{IC}$, so the remaining blocks are zero:

$$\mathbf{J}_{1,9} = \mathbf{J}_{2,8} = \mathbf{0}. \tag{B23}$$

On the DC side, the partial derivatives of the IC's DC-side injection (18) of the paper with respect to $\mathbf{x}_{IC}$, obtained from (B12), give $\mathbf{J}_{3,8}$ and $\mathbf{J}_{3,9}$, whose (k,c)th elements are defined as:

$$\begin{aligned} \mathbf{J}_{3,8}(k,c) &= \begin{cases} -1 - \big(b_c + 2c_c|I_{IC,c}|\big)\dfrac{\tilde{P}_{IC,AC,c}}{|I_{IC,c}|\,|V_{\mathrm{conv},c}|^2}, & c \in \mathcal{C}_k \\ 0, & \text{otherwise} \end{cases} \\ \mathbf{J}_{3,9}(k,c) &= \begin{cases} -\big(b_c + 2c_c|I_{IC,c}|\big)\dfrac{\tilde{Q}_{IC,AC,c}}{|I_{IC,c}|\,|V_{\mathrm{conv},c}|^2}, & c \in \mathcal{C}_k \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \tag{B24}$$

The internal circuit of IC c is represented by the transformer admittance $y_{t,c} = g_{t,c} + \mathrm{j}b_{t,c}$, the reactor admittance $y_{r,c} = g_{r,c} + \mathrm{j}b_{r,c}$, and the filter susceptance $B_{f,c}$ of (13) of the paper. Here pcc, flt, and conv denote the PCC, filter, and converter buses

of IC c. Their voltage magnitudes are V_{pcc}, V_{flt}, and V_{conv}, and their angles δ_{pcc}, δ_{flt}, and δ_{conv}. These magnitudes correspond to $|V_{\mathrm{pcc},c}|$, $|V_{\mathrm{flt},c}|$, and $|V_{\mathrm{conv},c}|$ in the paper. Among the AC buses, the IC constraints depend only on the internal-circuit bus voltages.

The partial derivatives of (36)–(37) of the paper with respect to the internal-circuit bus voltage magnitudes give $\mathbf{J}_{7,1}$ and $\mathbf{J}_{8,1}$, whose (c,i)th elements are defined as:

$$\mathbf{J}_{7,1}(c,i)=\begin{cases}-2g_{t,c}\big(V_{\mathrm{pcc}}-V_{\mathrm{flt}}\cos(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})\big), & i=\mathrm{pcc},\ \text{within the limit}\\ -2g_{t,c}\big(V_{\mathrm{flt}}-V_{\mathrm{pcc}}\cos(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})\big) & i=\mathrm{flt},\ \text{within the limit}\\ \quad -2g_{r,c}\big(V_{\mathrm{flt}}-V_{\mathrm{conv}}\cos(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}})\big), & \\ -2g_{r,c}\big(V_{\mathrm{conv}}-V_{\mathrm{flt}}\cos(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}})\big), & i=\mathrm{conv},\ \text{within the limit}\\ -\operatorname{sign}(\cdot)\,I_{IC,c}^{\max}, & i=\mathrm{conv},\ \text{at the limit},\\ & S_{IC,c}^{\max}=I_{IC,c}^{\max}V_{\mathrm{conv}}\\ 0, & \text{otherwise}\end{cases}\tag{B25}$$

$$\mathbf{J}_{8,1}(c,i)=\begin{cases}m_{qIC,c}+2b_{t,c}\big(V_{\mathrm{pcc}}-V_{\mathrm{flt}}\cos(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})\big), & i=\mathrm{pcc},\ \text{within the limit}\\ 2b_{t,c}\big(V_{\mathrm{flt}}-V_{\mathrm{pcc}}\cos(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})\big) & i=\mathrm{flt},\ \text{within the limit}\\ \quad +2b_{r,c}\big(V_{\mathrm{flt}}-V_{\mathrm{conv}}\cos(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}})\big)+2B_{f,c}V_{\mathrm{flt}}, & \\ 2b_{r,c}\big(V_{\mathrm{conv}}-V_{\mathrm{flt}}\cos(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}})\big), & i=\mathrm{conv},\ \text{within the limit}\\ -\operatorname{sign}(\cdot)\dfrac{\big(I_{IC,c}^{\max}\big)^2V_{\mathrm{conv}}}{Q_{IC,c}^{\max}}, & i=\mathrm{conv},\ \text{at the limit},\\ & S_{IC,c}^{\max}=I_{IC,c}^{\max}V_{\mathrm{conv}},\\ & Q_{IC,c}^{\max}>0\\ 0, & \text{otherwise}\end{cases}\tag{B26}$$

The partial derivatives of (36)–(37) of the paper with respect to the internal-circuit bus voltage angles give $\mathbf{J}_{7,3}$ and $\mathbf{J}_{8,3}$, whose (c,i)th elements are defined as:

$$\mathbf{J}_{7,3}(c,i)=\begin{cases}-2g_{t,c}V_{\mathrm{pcc}}V_{\mathrm{flt}}\sin(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}}), & i=\mathrm{pcc},\ \text{within the limit}\\ \ \ 2g_{t,c}V_{\mathrm{pcc}}V_{\mathrm{flt}}\sin(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})-2g_{r,c}V_{\mathrm{flt}}V_{\mathrm{conv}}\sin(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}}), & i=\mathrm{flt},\ \text{within the limit}\\ \ \ 2g_{r,c}V_{\mathrm{flt}}V_{\mathrm{conv}}\sin(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}}), & i=\mathrm{conv},\ \text{within the limit}\\ 0, & \text{otherwise}\end{cases}\tag{B27}$$

$$\mathbf{J}_{8,3}(c,i)=\begin{cases}\ \ 2b_{t,c}V_{\mathrm{pcc}}V_{\mathrm{flt}}\sin(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}}), & i=\mathrm{pcc},\ \text{within the limit}\\ -2b_{t,c}V_{\mathrm{pcc}}V_{\mathrm{flt}}\sin(\delta_{\mathrm{pcc}}-\delta_{\mathrm{flt}})+2b_{r,c}V_{\mathrm{flt}}V_{\mathrm{conv}}\sin(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}}), & i=\mathrm{flt},\ \text{within the limit}\\ -2b_{r,c}V_{\mathrm{flt}}V_{\mathrm{conv}}\sin(\delta_{\mathrm{flt}}-\delta_{\mathrm{conv}}), & i=\mathrm{conv},\ \text{within the limit}\\ 0, & \text{otherwise}\end{cases}\tag{B28}$$

The partial derivatives of (36)–(37) of the paper with respect to ω give $\mathbf{J}_{7,2}$ and $\mathbf{J}_{8,2}$, whose cth elements are defined as:

$$\begin{aligned}\mathbf{J}_{7,2}(c)&=\begin{cases}\dfrac{2m_{\omega,c}}{\omega_{\max}-\omega_{\min}}-\dfrac{\partial g_{t,c}}{\partial\omega}\big|V_{\mathrm{pcc}}e^{\mathrm{j}\delta_{\mathrm{pcc}}}-V_{\mathrm{flt}}e^{\mathrm{j}\delta_{\mathrm{flt}}}\big|^2-\dfrac{\partial g_{r,c}}{\partial\omega}\big|V_{\mathrm{flt}}e^{\mathrm{j}\delta_{\mathrm{flt}}}-V_{\mathrm{conv}}e^{\mathrm{j}\delta_{\mathrm{conv}}}\big|^2, & \text{within the limit}\\ 0, & \text{at the limit}\end{cases}\\ \mathbf{J}_{8,2}(c)&=\begin{cases}\dfrac{\partial b_{t,c}}{\partial\omega}\big|V_{\mathrm{pcc}}e^{\mathrm{j}\delta_{\mathrm{pcc}}}-V_{\mathrm{flt}}e^{\mathrm{j}\delta_{\mathrm{flt}}}\big|^2+\dfrac{\partial b_{r,c}}{\partial\omega}\big|V_{\mathrm{flt}}e^{\mathrm{j}\delta_{\mathrm{flt}}}-V_{\mathrm{conv}}e^{\mathrm{j}\delta_{\mathrm{conv}}}\big|^2+\dfrac{B_{f,c}}{\omega}V_{\mathrm{flt}}^2, & \text{within the limit}\\ 0, & \text{at the limit}\end{cases}\end{aligned}\tag{B29}$$

Specifically, the derivatives of $g_{t,c}$, $g_{r,c}$, $b_{t,c}$, and $b_{r,c}$ follow from (1) of the paper:

$$\begin{aligned}\frac{\partial g_{t,c}}{\partial\omega}&=\frac{-2R_{t,c}X_{t,c}^2\omega}{\big(R_{t,c}^2+X_{t,c}^2\omega^2\big)^2},\quad \frac{\partial b_{t,c}}{\partial\omega}=\frac{-X_{t,c}\big(R_{t,c}^2-X_{t,c}^2\omega^2\big)}{\big(R_{t,c}^2+X_{t,c}^2\omega^2\big)^2},\\ \frac{\partial g_{r,c}}{\partial\omega}&=\frac{-2R_{r,c}X_{r,c}^2\omega}{\big(R_{r,c}^2+X_{r,c}^2\omega^2\big)^2},\quad \frac{\partial b_{r,c}}{\partial\omega}=\frac{-X_{r,c}\big(R_{r,c}^2-X_{r,c}^2\omega^2\big)}{\big(R_{r,c}^2+X_{r,c}^2\omega^2\big)^2},\end{aligned}\tag{B30}$$

where $R_{t,c}$ and $X_{t,c}$ denote the resistance and reactance of the transformer branch, and $R_{r,c}$ and $X_{r,c}$ those of the reactor branch, so that the reactance in (1) of the paper is $\omega X_{t,c}$ and $\omega X_{r,c}$. The DC bus voltage enters (36) through $P_{IC,AC,c}$ of (9)

and does not appear in (37) of the paper. The partial derivatives of (36)–(37) of the paper with respect to $\mathbf{V}_{DC}$ give $\mathbf{J}_{7,4}$ and $\mathbf{J}_{8,4}$, whose (c,k)th elements are defined as:

$$\mathbf{J}_{7,4}(c,k) = \begin{cases} -\dfrac{2m_{V_{DC},c}}{V_{DC,\max,c} - V_{DC,\min,c}}, & c \in \mathcal{C}_k, \text{ within the limit} \\ 0, & \text{otherwise} \end{cases} \tag{B31}$$

$$\mathbf{J}_{8,4} = \mathbf{0}.$$

The reactive-power limit $Q_{IC,c}^{\max}$ in (20) of the paper varies with the active-power injection. The partial derivatives of (37) of the paper with respect to $\tilde{\mathbf{P}}_{IC}$ give $\mathbf{J}_{8,8}$, whose (c,c')th elements are defined as:

$$\mathbf{J}_{8,8}(c,c') = \begin{cases} \text{sign}(\cdot)\dfrac{\tilde{P}_{IC,AC,c}}{Q_{IC,c}^{\max}}, & c' = c, \text{ at the limit, } Q_{IC,c}^{\max} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{B32}$$

The two constraint groups $\mathbf{F}_{P,IC}$ and $\mathbf{F}_{Q,IC}$ contain $\tilde{\mathbf{P}}_{IC}$ and $\tilde{\mathbf{Q}}_{IC}$ themselves, and no remaining term of each constraint depends on its own variable. The blocks $\mathbf{J}_{7,8}$ and $\mathbf{J}_{8,9}$ are again identity matrices:

$$\mathbf{J}_{7,8} = \mathbf{J}_{8,9} = \mathbf{I}. \tag{B33}$$

The constraint $\mathbf{F}_{P,IC}$ does not contain $\tilde{\mathbf{Q}}_{IC}$, so the remaining block is zero:

$$\mathbf{J}_{7,9} = \mathbf{0}. \tag{B34}$$

Each constraint contains none of the generator variables $\tilde{\mathbf{P}}_{G,AC}$, $\tilde{\mathbf{Q}}_{G,AC}$, and $\tilde{\mathbf{P}}_{G,DC}$ either, so the corresponding blocks are zero:

$$\begin{aligned} \mathbf{J}_{7,5} = \mathbf{J}_{7,6} = \mathbf{J}_{7,7} = \mathbf{0}, \\ \mathbf{J}_{8,5} = \mathbf{J}_{8,6} = \mathbf{J}_{8,7} = \mathbf{0}. \end{aligned} \tag{B35}$$

D. Changes in the Structure of the Augmented Jacobian

TABLE B2
ROWS AND COLUMNS REMOVED UNDER EACH OPERATING MODE

Generators		
Operating Mode	Removed Column	Removed Row
AC Side		
Slack	ω, $\lvert V_i\rvert$	$F_{P,AC,g}$, $F_{Q,AC,g}$
PV	$\lvert V_i\rvert$	$F_{Q,AC,g}$
PQ	—	—
Droop	—	—
DC Side		
CV_{DC}	V_k	$F_{P,DC,g}$
CP	—	—
Droop	—	—

ICs and DC/DC Converters		
Operating Mode	Removed Column	Removed Row
IC Active Power Control		
CV_{DC}	V_k	$F_{P,IC,c}$
CF	ω	$F_{P,IC,c}$
CP	—	—
Droop	—	—
IC Reactive Power Control		
CV_{AC}	$\lvert V_i\rvert$	$F_{Q,IC,c}$
CQ	—	—
Droop	—	—
DC/DC Converter		
CV_{HV} $(d \in \mathcal{D}_k^{HV})$	V_k	$\Delta P_{DC,k}$
CV_{LV} $(d \in \mathcal{D}_k^{LV})$	V_k	$\Delta P_{DC,k}$
CP	—	—
Droop	—	—

The structure of the augmented Jacobian changes according to the operating mode of each generator, IC, and DC/DC converter and to whether it reaches an output limit. As in Section III-C of the paper, the column of a state variable fixed by setting a droop coefficient in (6)–(10) and (22) to $m \approx \infty$ is removed from the augmented Jacobian, since the variable does not enter the unified state vector as an unknown. The augmented constraint row of the corresponding generator output is removed, since the output is determined by the power balance rather than its characteristic. For an IC, the augmented constraint row of the corresponding converter-bus injection is removed in the same manner, since the converter-bus injection is also determined by the power balance rather than its characteristic. Since the active-power injection of a DC/DC converter enters the specified injection (29) of the paper directly through (22) and (23), the mismatch row at the bus whose voltage the converter fixes is removed instead.

Table B2 summarizes the state variable fixed by each operating mode and the column and row removed accordingly. For generators, the slack mode on the AC side fixes both ω and $|V_i|$, and removes the augmented constraint rows $F_{P,AC,g}$ and $F_{Q,AC,g}$, whereas the PV mode fixes only $|V_i|$ and removes $F_{Q,AC,g}$. On the DC side, the CV_{DC} mode of a generator fixes V_k and removes $F_{P,DC,g}$. For converters, the CV_{DC} and CF modes for the active-power control of an IC fix V_k and ω, respectively, and remove the augmented constraint row $F_{P,IC,c}$. The CV_{AC} mode for the reactive-power control of an IC fixes $|V_i|$ and removes the augmented constraint row $F_{Q,IC,c}$. The CV_{HV} and CV_{LV} modes of a DC/DC converter whose high- and low-voltage bus is bus k, respectively, fix V_k and remove $\Delta P_{DC,k}$. The active-power injection of the converter determined by the power balance at bus k then enters the mismatch row at the other bus through (23) of the paper. The mismatch at bus k', the high-voltage bus of the converter in the CV_{LV} mode and its low-voltage bus in the CV_{HV} mode, therefore depends on the voltages of the DC buses connected to bus k through the power balance at bus k, whereas (B11) contains only the derivatives of the droop terms in (22) of the paper. To account for this dependence, substituting the active-power injection of the converter determined at bus k into the mismatch at bus k' through (23) of the paper gives the mismatch at bus k' as:

$$\Delta \bar{P}_{DC,k'} = \Delta P_{DC,k'} + \Delta P_{DC,k}. \tag{B36}$$

Accordingly, the k'th row of each block $\mathbf{J}_{3,b}$ in Table B1 becomes the sum of its k'th and kth rows as:

$$\bar{\mathbf{J}}_{3,b}(k',\cdot) = \mathbf{J}_{3,b}(k',\cdot) + \mathbf{J}_{3,b}(k,\cdot), \qquad b = 1,\ldots,9. \tag{B37}$$

Since $P_{conv,LV,d}$ and $P_{conv,HV,d}$ cancel in (B36) by (23) of the paper, the k'th row of $\mathbf{J}_{3,4}$ in (B37) contains $-\partial P_{cal,DC,k}/\partial V_l$ for each DC bus l connected to bus k. Neither a column nor a row is removed for the remaining PQ, CP, CQ, and droop modes, since they fix no state variable.

As shown in Table B2, more than one device can fix the same state variable. Since the column of that state variable is removed from the augmented Jacobian only once while each device removes a row of its own, only one device must be set to regulate it. On the AC side, $|V_i|$ can be fixed by a generator in the slack or PV mode or an IC in the CV_{AC} mode at the same bus, and ω by a generator in the slack mode or an IC in the CF mode, with only one device set to regulate each variable. On the DC side, V_k can be fixed by a generator in the CV_{DC} mode, an IC in the CV_{DC} mode, or a DC/DC converter in the CV_{HV} or CV_{LV} mode at the same bus, with only one device set to regulate it.

In addition, a generator or IC that reaches an output limit holds its output at that limit rather than the state variable of its mode, which remains unknown. The column and row that its operating mode removes are therefore restored, as in the PQ, CP, and CQ modes of Table B2. For generators, the column and augmented constraint row are restored at the reactive-power limit in the PV mode on the AC side, and at the active-power limit in the CV_{DC} mode on the DC side. For ICs, the column and augmented constraint row are restored in the CV_{DC} and CF modes when the active-power injection reaches $P^{\max}_{IC,c}$, and in the CV_{AC} mode at the reactive-power limit. When a generator or IC operating in a droop mode reaches an output limit, neither a column nor a row is restored, since the droop modes fix no state variable.

If the DC/DC converter model includes the conversion loss and output limits, the active-power injection of the converter is carried as an augmented variable in the same manner as for an IC in (35) of the paper. The constant-voltage modes of the DC/DC converter then remove the augmented constraint row rather than the mismatch row $\Delta P_{DC,k}$, which is retained and instead determines the active-power injection.

APPENDIX C
ADDITIONAL CASE STUDIES

A. Three-Winding Transformer Validation

The three-winding transformer model is validated on an AC/DC test system modified from pandapower's multi-voltage example network [2], [3], shown in Fig. C1. The three-winding transformer of the 24-bus AC network connects the 110-kV, 20-kV, and 10-kV levels, while two-winding transformers link the 380-kV source to the 110-kV level and the 10-kV level to the 0.4-kV feeders. A seven-bus 0.75-kV DC grid is connected to the 0.4-kV feeders through three ICs. For active-power control, IC1 operates in the CV_{DC} mode while IC2 and IC3 operate in the CP mode, and for reactive-power control all three ICs operate in the CQ mode.

Fig. C2 shows the voltage magnitudes at every AC and DC bus, together with the AC-side voltage angles, for both the proposed and the pandapower solutions. As shown in Fig. C2, the solutions of the proposed formulation are almost identical to the pandapower reference. The maximum deviations are 7.8×10^{-7} p.u. in voltage magnitude and 8.9×10^{-5} deg in angle over the AC buses, and 2.8×10^{-8} p.u. over the DC buses. The three-winding transformer model is therefore validated.

B. Large AC/DC Transmission System

The proposed formulation is next applied to a modified 2869-bus PEGASE model of the European transmission system [4], [5], shown in Fig. C3, and Table C1 summarizes the test system. In this system, a three-terminal and a four-terminal meshed MTDC grid are connected to the 380-kV transmission network through seven ICs. All seven ICs operate in P–V_{DC} droop mode,

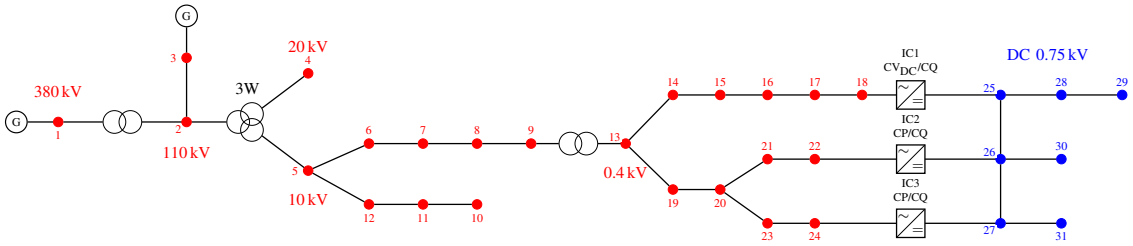


Fig. C1. AC/DC test system modified from pandapower's multi-voltage example network.

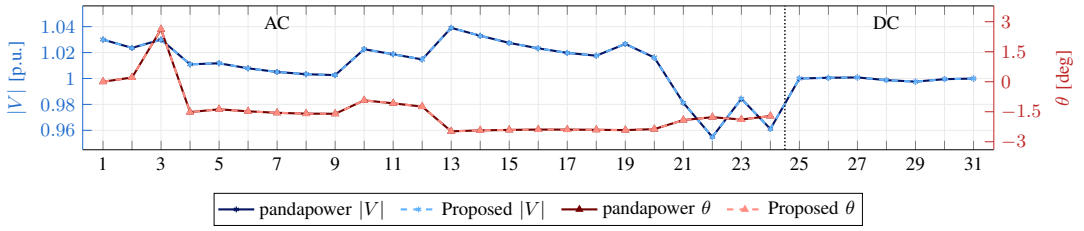


Fig. C2. Comparison of the bus voltage magnitudes and the AC-side voltage angles with the pandapower solution on the AC/DC test system.

each reaching its rating for a 0.5 % change in DC voltage. The proposed solution is then compared with MatACDC [6], which solves the AC and DC subgrids through a sequential outer-loop procedure. As shown in Table C1, the maximum deviations between the proposed and MatACDC results are 1.3×10^{-6} p.u. in voltage magnitude and 2.7×10^{-3} deg in angle over the 2869 AC buses, and 1.9×10^{-5} p.u. over the seven DC buses.

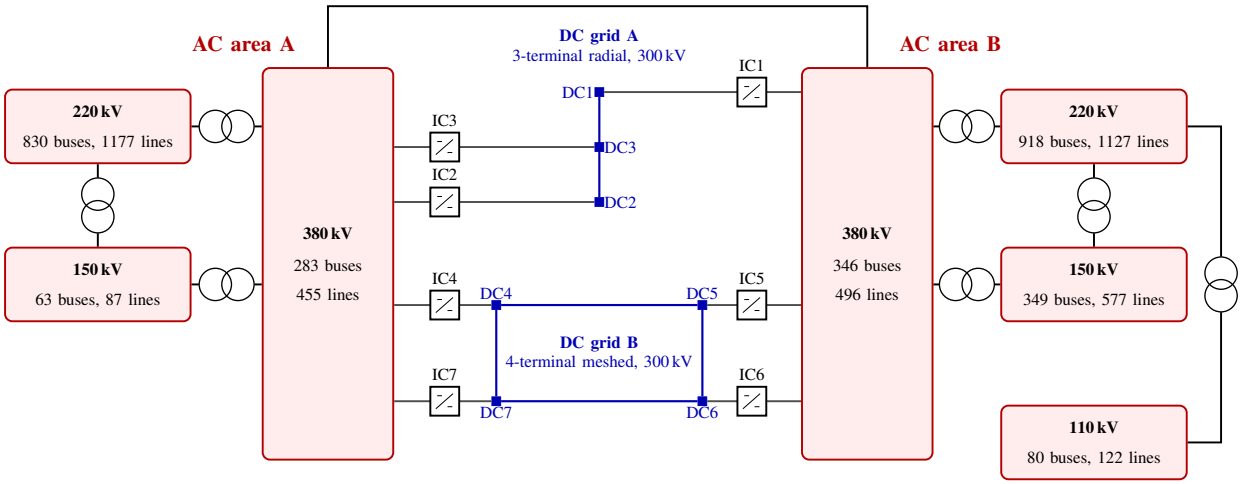


Fig. C3. Modified 2869-bus PEGASE AC/DC hybrid test system.

TABLE C1
MODIFIED 2869-BUS PEGASE AC/DC HYBRID TEST SYSTEM AND COMPARISON WITH MATACDC

Test system	
AC network	2869 buses, 4582 branches, 510 generators
DC network	7 buses, 6 lines, 300 kV monopolar
Interlinking converters	7 units, all in P–V_{DC} droop/CQ mode
DC grids	3-terminal radial and 4-terminal meshed
Deviation between the proposed and MatACDC solutions	
$\max \lvert\Delta \lvert V\rvert\rvert$ over 2869 AC buses [p.u.]	1.3×10^{-6}
$\max \lvert\Delta\theta\rvert$ over 2869 AC buses [deg]	2.7×10^{-3}
$\max \lvert\Delta V_{DC}\rvert$ over 7 DC buses [p.u.]	1.9×10^{-5}

Beyond this validation, convergence is tested under strong AC/DC coupling, as in Section IV-A of the paper. Table C2 summarizes the comparison of the iteration counts and computational times between the proposed formulation and the sequential

method as the active-power set-points of the seven ICs are scaled up. As shown in Table C2, the proposed formulation still converges at a scaling factor of 30, where the sequential method fails. In addition, the proposed formulation requires approximately one-third of the computational time of the sequential method. The proposed formulation therefore remains applicable to an AC/DC transmission network with 2869 AC buses.

TABLE C2
CONVERGENCE AND COMPUTATIONAL TIME AS THE POWER EXCHANGE THROUGH ALL SEVEN ICS IS SCALED

	Proposed (unified)		**MatACDC** (sequential)		
Scaling factor	Iterations	Time [ms]	Outer iterations	AC NR iterations	Time [ms]
×1	6	160.5	3	10	513.8
×18	6	159.9	6	22	626.1
×19	6	162.5		no convergence	
×30	6	159.0		no convergence	

C. Convergence Comparison with an Outer-Loop Implementation

The proposed formulation is finally compared with an outer-loop implementation on the system in Section IV-C of the paper, in terms of convergence and computational time. None of the existing methods summarized in Table V of the paper implements every outer-loop element. The outer-loop elements are therefore implemented by adding an outer loop to the baseline NR formulation, as shown in Fig. 3(a) of the paper, whereas the proposed formulation resolves them within the unified NR iteration, as shown in Fig. 3(b). Among the outer-loop elements, the generator droop deadband in Scenario C2 is examined as the loads are scaled.

Table C3 summarizes the convergence and computational time as the loads are scaled. Each computational time is the average of 150 runs. The proposed formulation requires approximately three-fifths of the computational time of the outer-loop implementation. At a scaling factor of 1.50, the proposed formulation still converges, whereas the outer-loop implementation fails to converge. The reason the outer-loop implementation fails to converge under heavy loading is that the operating modes are updated only outside the NR iteration. As a result, every generator output remains fixed at its set-point inside the deadband during the NR iteration and provides no voltage support. Compared with the outer-loop implementation, the proposed formulation therefore has advantages in terms of both convergence and computational time.

TABLE C3
CONVERGENCE AND COMPUTATIONAL TIME AS THE LOADS ARE SCALED

	Proposed (unified loop)		**Outer-loop implementation**		
Scaling factor	Iterations	NR time [ms]	Outer iterations	NR iterations	NR time [ms]
×1.00	4	4.3	2	4	6.3
×1.45	7	4.9	3	11	9.1
×1.50	7	5.1		no convergence	
×1.55	7	5.0		no convergence	

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